

Variables and parameters

V6_production.tex

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The classification is for the **two-country general equilibrium**, including the specialized technology and bubble analysis. A quantity taken as given by one household or firm need not be exogenous in general equilibrium. Country and type variants are grouped where their economic roles coincide; branch superscripts and Markov-function versions inherit the classification of the underlying quantity. Proof notation is included separately within the relevant category so that it is not mistaken for an additional calibration input.

Category	Classification rule
Parameters	Fixed preference, technology, demographic, and transition-law inputs. Composite parameters and mathematical constants used in proofs are identified explicitly; they are not additional independent structural parameters.
Exogenous variables	Regime realizations and externally supplied initial conditions. Initial knowledge is supplied to the model, whereas subsequent knowledge is an endogenous state.
Endogenous variables	Equilibrium states, allocations, prices, holdings, returns, and quantities calculated from them. This category includes equilibrium policy functions, accounting identities, and selected balanced-growth values.

Two classification choices are consequential. First, the manuscript defines $Q_{i,t} = N_{i,t+1}q_{i,t}$, while $\mathcal{D}_{i,t} = N_{i,t}d_{i,t}$: capitalization values the post-issuance stock supply at the current ex-dividend price. Second, the productivity **maps** $A_{X,i}(\cdot), A_{L,i}(\cdot)$ are primitive functions, but their realized **levels** depend on endogenous knowledge and the exogenous regime. They are therefore listed as derived endogenous quantities, not as independently specified productivity paths.

1 Parameters

1.1 Primitive scalar and functional parameters

Symbol	Economic meaning	Definition, restriction, or relationship to other objects
$H_i, i \in \{US, W\}$	Mass of skilled young agents born each date	Fixed, strictly positive; supplies skilled labor to intermediate production and R&D.
L_i	Mass of unskilled young agents born each date	Fixed, strictly positive; supplies labor to final-good production. A fixed endowment parameter, distinct from leakage L_t^D, L_t^S .
a_i	R&D productivity per skilled worker per existing variety	$a_i > 0$; one R&D worker creates $a_i N_{i,t}$ new varieties.
ϑ_i	Curvature of the intermediate-variety aggregator; inverse monopoly markup	$0 < \vartheta_i < 1$; demand elasticity is $1/(1 - \vartheta_i)$ and price/wage markup is $1/\vartheta_i$.
$F_i(\cdot, \cdot)$	Country final-good production technology	Primitive function: continuously differentiable, concave, homogeneous of degree one, with positive partial derivatives. Only the US function is subsequently specialized to CES; the RoW function remains general.

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Symbol	Economic meaning	Definition, restriction, or relationship to other objects
$A_{X,i}(\cdot), A_{L,i}(\cdot)$	Primitive state-to-productivity maps	Measurable functions of $s = (z, N_{US}, N_W)$. Their functional forms are inputs; evaluating them at an endogenous state produces the levels in Table 3.1.
β	Weight on old-age utility	$0 < \beta < 1$; US saving rate is β , while the RoW saving rate is $(\beta + \chi)/(1 + \chi)$.
γ	Risk-aversion parameter in the Epstein–Zin aggregator	General portfolio block: $\gamma > 0$, $\gamma \neq 1$ in the displayed formula. The sufficient bubble theorem additionally requires $0 < \gamma < 1$.
κ	Intensity of the quadratic equity-share preference penalty	Utility contains $-\kappa(\omega - \bar{\omega})^2/2$, with the same coefficient for RoW. The source does not explicitly state a global sign/domain restriction; a nonnegative penalty interpretation is not a separately written assumption.
$\bar{\omega}$	US household target weight on US equity within its equity portfolio	Explicitly restricted to $[1/2, 1]$. Distinct from the endogenous actual weight ω_t .
$\bar{\omega}^*$	RoW household target weight on US equity within its equity portfolio	Appears in $-\kappa(\omega_{a,t}^* - \bar{\omega}^*)^2/2$. The source gives no explicit interval for this target; no restriction such as $\bar{\omega}^* \leq 1/2$ is imposed here.
η	Intensity of the US bond-share utility cost	Multiplies $\Upsilon(\theta)$; the proof explicitly permits $\eta = 0$. A full global domain is not separately stated in the source.
$\Upsilon(\cdot)$	US bond-share cost function	Primitive differentiable function on $(-\infty, 1)$, with Υ' bounded near zero. No global coercivity condition is imposed. $\Upsilon'(\theta_t)$ is its derivative evaluated at an endogenous share.
χ	RoW convenience-utility coefficient for US bonds	$\chi > 0$; utility includes $\chi \log(q_{US,t}^B B_{US,a,t+1}^*)$.
Π	Markov transition matrix for the regime	Primitive probabilities $\Pi(z, z')$. In the specialized two-state model, ordered as (u, b) , $\Pi = \begin{pmatrix} \pi & 1 - \pi \\ 0 & 1 \end{pmatrix}$. Thus Π and π are not independent inputs after specialization.
π	Probability of remaining in regime u for one more period	$\pi \in (0, 1)$; the switch probability is $1 - \pi$, and b is absorbing.
α_{US}	US CES distribution parameter	$0 < \alpha_{US} < 1$; $F_{US}(X, L) = [\alpha_{US} X^{1-\rho_{US}} + (1 - \alpha_{US}) L^{1-\rho_{US}}]^{1/(1-\rho_{US})}$.
ρ_{US}	US CES curvature parameter	$\rho_{US} > 1$ in the specialized model. In this parameterization the elasticity of substitution is $1/\rho_{US}$, not ρ_{US} .
$\bar{A}_{X,US,u}, \bar{A}_{L,US,u}$	US productivity scale coefficients in regime u	Both positive; $A_{X,US,t}^u = \bar{A}_{X,US,u} (N_{US,t}^u)^{\xi_u}$ and $A_{L,US,t}^u = \bar{A}_{L,US,u} (N_{US,t}^u)^{\nu_u}$.
$\bar{A}_{X,US,b}, \bar{A}_{L,US,b}$	US productivity scale coefficients in regime b	Both positive; both factor augmentations have knowledge exponent ν_b .
$\bar{A}_{X,W}, \bar{A}_{L,W}$	RoW productivity scale coefficients	Both positive and regime-invariant; both factor augmentations have knowledge exponent ξ_W .
ξ_u	Knowledge elasticity of US knowledge-input productivity in regime u	Exponent in $A_{X,US}^u = \bar{A}_{X,US,u} N_{US}^{\xi_u}$; specialized restrictions impose $\xi_u > \nu_u$.

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Symbol	Economic meaning	Definition, restriction, or relationship to other objects
ν_u	Knowledge elasticity of US unskilled-labor productivity in regime u	Exponent in $A_{L,US}^u = \bar{A}_{L,US,u} N_{US}^{\nu_u}$; $\nu_u > \nu_b$.
ν_b	Common knowledge elasticity of both US factor augmentations in regime b	$\nu_b \geq 0$ and $\xi_u > \nu_u > \nu_b$ in the sufficient-conditions technology block.
ξ_W	Common knowledge elasticity of both RoW factor augmentations	$\xi_W \geq 0$. Regime-invariance of these primitives does not make RoW allocations or prices regime-invariant.

1.2 Composite parameters and technology functions

These objects are determined by primitive parameters. They should be computed rather than independently calibrated. Constants that summarize a selected equilibrium, even when a closed-form expression is available, are reported with endogenous quantities in Table 3.7.

Symbol	Meaning	Relationship to more primitive parameters
ψ_{US}	Exponent governing US relative factor-productivity growth in the CES wage-bill relation	$\psi_{US} = (\xi_u - \nu_u)(\rho_{US} - 1) > 0$.
K_u	Coefficient in the exact US CES wage-bill ratio	$K_u = \frac{1 - \alpha_{US}}{\vartheta_{US} \alpha_{US}} \left(\frac{\bar{A}_{X,US,u} H_{US}}{\bar{A}_{L,US,u} L_{US}} \right)^{\rho_{US} - 1}$.
D_u	Constant in the limiting HKT scalar equation	$D_u = 1/(a_{US} H_{US}) + 1 - \beta > 0$. This is not a dividend.
G_u	Maximum US knowledge-growth factor; attained asymptotically on the selected all- u tail	$G_u = 1 + a_{US} H_{US}$. The finite-date factor $G_t = 1 + a_{US}(1 - \varphi_t) H_{US}$ is endogenous.
κ_D	Dividend-leakage decay exponent	$\kappa_D = \psi_{US}/\rho_{US} > 0$. Unrelated to the portfolio penalty κ .
κ_S	Switch-leakage decay exponent associated with US knowledge	$\kappa_S = (\nu_u - \nu_b)(1 - \gamma) > 0$ under the bubble theorem.
$B(x)$	CES output function after dividing by the augmented unskilled-labor input	$B(x) = [\alpha_{US} x^{1-\rho_{US}} + 1 - \alpha_{US}]^{1/(1-\rho_{US})}$. A technology function; its argument x_t is endogenous. Distinct from bonds and bubbles.
$\hat{w}_{H,W}(\varphi), \hat{w}_{L,W}(\varphi)$	Normalized RoW wage schedules	$\hat{w}_{H,W}(\varphi) = \vartheta_W F_{W,X}(\bar{A}_{X,W} \varphi H_W, \bar{A}_{L,W} L_W) \bar{A}_{X,W}$; $\hat{w}_{L,W}(\varphi) = F_{W,L}(\bar{A}_{X,W} \varphi H_W, \bar{A}_{L,W} L_W) \bar{A}_{L,W}$. The schedules follow from primitives; their equilibrium evaluations depend on $\varphi_{W,t}$.
$\hat{e}_W(\varphi)$	Normalized RoW young-income schedule	$\hat{e}_W(\varphi) = H_W \hat{w}_{H,W}(\varphi) + L_W \hat{w}_{L,W}(\varphi)$. Its supremum and that of $\hat{w}_{H,W}$ are assumed finite.

1.3 Auxiliary constants, thresholds, and proof constructions

The following are **mathematical parameters or derived bounds**, not additional economic primitives. Where a constant bounds an equilibrium path, that bound must be established or imposed as a selection condition; naming it does not make the underlying equilibrium variable exogenous.

Symbol	Role	Definition or relationship
c_U	Positive lower-bound coefficient for US income	$e_{US,t}^u \geq c_U N_t^{\nu_u}$. The proof constructs a uniform constant using primitives and $N_t \geq N_0 > 0$.
C_W	Upper-bound coefficient for RoW income	$e_{W,t}^u \leq C_W (N_{W,t}^u)^{\xi_w}$, using the bounded normalized income schedule.
C_{QW}, C_{DW}	Upper-bound coefficients for RoW capitalization and dividends	$\mathcal{Q}_{W,t}^u \leq C_{QW} (N_{W,t}^u)^{\xi_w}$ and $\mathcal{D}_{W,t}^u \leq C_{DW} (N_{W,t}^u)^{\xi_w}$.
C_{PW}	Uniform coefficient bounding the RoW successor equity payoff	$N_{W,t}(q_{W,t}^z + d_{W,t}^z) \leq C_{PW} N_{W,t}^{\xi_w}$ for $z \in \{u, b\}$.
C_ζ	Coefficient linking the funding deviation to relative country scale	$ \zeta_t - 1 \leq C_\zeta r_t$. Follows from country-scale bounds and world capitalization clearing.
N^I, ε^I	Thresholds sufficient for active US innovation	$N_t \geq N^I$ and $r_t \leq \varepsilon^I$ imply $\varphi_t < 1$; these thresholds are constructed from primitives.
$\bar{\varphi}_H$	Chosen upper bound on the production share inside the HKT cone	Chosen in $(0, 1)$ small enough that $\Gamma_H < 1$. It is not an equilibrium constant production share.
Γ_H	Upper bound on relative-scale growth inside the HKT cone	$\Gamma_H = \frac{(1 + a_W H_W)^{\xi_w}}{[1 + a_{US}(1 - \bar{\varphi}_H)H_{US}]^{\nu_u}} < 1$.
$\varepsilon_H, N^\dagger$	Relative-scale and US-knowledge thresholds defining the HKT cone	Chosen with $\varepsilon_H \leq \varepsilon^I$, $C_\zeta \varepsilon_H \leq 1/2$, and $N^\dagger \geq N^I$ sufficiently large to bound φ_t .
$\mathcal{C}_H$	Constructed forward-invariant set of knowledge stocks	$\mathcal{C}_H = \{(N, M) \in \mathbb{R}_{++}^2 : N \geq N^\dagger, M^{\xi_w}/N^{\nu_u} \leq \varepsilon_H\}$. Here N, M are set coordinates, not pricing kernels.
C_φ	Coefficient in the production-share upper bound	$C_\varphi = \left[\frac{2(1/(a_{US}H_{US}) + 1)}{\beta K_u} \right]^{1/\rho_{US}}$; $\varphi_t \leq C_\varphi N_t^{-\psi_{US}/\rho_{US}}$ inside the cone.
$\hat{\varphi}$	Chosen production-share bound for the finite-prefix sufficient condition	Chosen in $(0, 1)$ to ensure $\Gamma_{\text{pre}} < 1$. Distinct from an observed or solved labor share.
$\Gamma_{\text{pre}}, \underline{G}_{US}$	Prefix relative-scale contraction bound and US knowledge-growth lower bound	$\underline{G}_{US} = 1 + a_{US}(1 - \hat{\varphi})H_{US}$; $\Gamma_{\text{pre}} = (1 + a_W H_W)^{\xi_w} / \underline{G}_{US}^{\nu_u} < 1$.
$\underline{\zeta}, \underline{\omega}$	Lower bounds imposed or verified on finite-prefix equilibrium funding and US equity weight	Require $\zeta_t \geq \underline{\zeta} > 0$; the sufficient portfolio condition is $\omega_t^u \geq \underline{\omega} > 0$. Under that condition one may set $\underline{\zeta} = \underline{\omega}$. These are selection/verification constants, not preference targets.

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Symbol	Role	Definition or relationship
T_N, T_r, T^*	Derived sufficient horizons for meeting the cone thresholds	$T_N = \lceil \log(N^\dagger/N_0)/\log \underline{G}_{US} \rceil_+$; $T_r = \lceil \log(r_0/\varepsilon_H)/(-\log \Gamma_{\text{pre}}) \rceil_+$; $T^* = \max\{T_N, T_r\}$. They are proof bounds calculated from thresholds and initial conditions, not independent horizon calibrations.
$C_{w,b}, C_b$	Uniform US switch-state wage and equity-payoff bounds	$w_{H,US,t}^b \leq C_{w,b} N_t^{\nu_b}$ and $q_t^b + d_t^b \leq C_b N_t^{\nu_b-1}$.
C_D, C_S	Coefficients bounding the two contributions to total leakage	$a_t \leq C_D N_t^{-\kappa_D} + \frac{1-\pi}{\pi} C_S [N_t^{-\kappa_S} + r_{t-1}^{1-\gamma} + r_{t-1}]$ eventually.
ϵ_0, N_0^H	Domain thresholds for existence of the interior scalar terminal root	$0 < \epsilon_0 < 1$; require $ \zeta - 1 \leq \epsilon_0$ and $N \geq N_0^H$. Choose $\beta(1 + \epsilon_0) < 1/(a_{US}H_{US}) + 1$. N_0^H is not initial US knowledge N_0 .
C_H	Bound on sensitivity of the scalar terminal root to funding	$ \log \varphi^{2C}(N_t, \zeta_t) - \log \varphi^{HKT}(N_t) \leq C_H \zeta_t - 1 $. Distinct from the set $\mathcal{C}_H$.
C	Generic finite bound used locally in proofs	A placeholder whose value can change across inequalities; it is not a single shared economic parameter.

2 Exogenous variables and initial conditions

Symbol	Role	Definition or relationship
z_t	Exogenous aggregate regime state	$z_t \in \{u, b\}$ with law II. In the specialized model $\Pr(z_{t+1} = u \mid z_t = u) = \pi$ and $\Pr(z_{t+1} = b \mid z_t = b) = 1$.
z_0	Supplied initial regime	Part of s_0 . The bubble analysis starts from $z_0 = u$.
$N_{US,0}, N_{W,0}; N_0 := N_{US,0}^u$	Supplied initial knowledge stocks	Strictly positive initial conditions. They are predetermined inputs for an equilibrium path; $N_{i,t}$ at later dates is endogenous through past R&D.
s_0	Supplied initial aggregate state	$s_0 = (z_0, N_{US,0}, N_{W,0}) \in \mathcal{S}$. This does not make the entire process s_t exogenous.
r_0	Relative real scale at the supplied initial state	$r_0 = (N_{W,0}^u)^{\xi_w} / N_0^{\nu_u}$; a derived initial condition, not an additional independent input.
τ	First-switch date; exogenous stopping time in the production application	$\tau = \min\{t \geq 1 : z_t = b\}$ when $z_0 = u$. Section 1 allows a general first-switch environment; the production specialization makes this a function of the exogenous regime history.
$\mathbf{1}_{\{\tau > t\}}, \mathbf{1}_{\{\tau = t\}}, \mathbf{1}_{\{\tau \leq T\}}, \mathbf{1}_{\{\tau < \infty\}}$	Indicators of survival, switching, or eventual switching	Functions of the exogenous regime path. In the specialized model, $\Pr(\tau > t) = \pi^t$ and $\Pr(\tau = t) = \pi^{t-1}(1 - \pi)$. The indicators are not additional shocks.

The model contains no independently specified labor-income/endowment process $e_{i,t}$, no exogenous risk-free-rate process, and no independent law of motion for the productivity levels once their state maps are given. Initial financial holdings of the young are fixed at zero by the model structure; no separate named initial-wealth state is introduced.

3 Endogenous variables

3.1 States, production, innovation, income, and equity values

Symbol	Economic meaning and role	Definition or relationship to more primitive quantities
$N_{i,t}; N_t := N_{US,t}^u$	Knowledge stock / measure of incumbent varieties; endogenous predetermined state	$N_{i,t+1} = [1 + a_i(1 - \varphi_{i,t})H_i]N_{i,t}$. At a given date it is inherited, but its path is determined by equilibrium R&D.
s_t, s	Aggregate state, containing endogenous and exogenous components	$s_t = (z_t, N_{US,t}, N_{W,t})$. Classified here as a mixed state vector, not as a wholly exogenous process.
$A_{X,i,t}, A_{L,i,t}; A_{X,i,t}^z, A_{L,i,t}^z$	Realized factor-augmenting productivities; technologically determined endogenous levels	$A_{X,i,t} = A_{X,i}(s_t)$ and $A_{L,i,t} = A_{L,i}(s_t)$. Under specialization: US u exponents (ξ_u, ν_u) , US b exponents (ν_b, ν_b) , and RoW exponents (ξ_W, ξ_W) , multiplied by the corresponding $\bar{A}$ coefficients.
$\varphi_{i,t}, \varphi_i(s); \varphi_t := \varphi_{US,t}^u$	Skilled-labor share in intermediate production; equilibrium allocation	$0 < \varphi_{i,t} \leq 1$. The R&D share is $1 - \varphi_{i,t}$; the production and research labor masses are $\varphi_{i,t}H_i$ and $(1 - \varphi_{i,t})H_i$. Determined jointly by free-entry complementarity and equilibrium funding.
$X_{i,t}, X_i(s)$	Quantity of the knowledge-intensive input	$X_{i,t} = N_{i,t}x_{i,t} = \varphi_{i,t}H_i$ in symmetric equilibrium; its primitive aggregator is the normalized continuum CES expression.
$x_{i,t}(j), x_{i,t}$	Output and skilled-labor input of an individual intermediate firm	Symmetry gives $x_{i,t}(j) = x_{i,t} = X_{i,t}/N_{i,t}$. Individual demand is $x_{i,t}(j) = (X_{i,t}/N_{i,t})[p_{i,t}(j)/P_{i,t}]^{-1/(1-\vartheta_i)}$.
$p_{i,t}(j)$; local choice p	Intermediate variety price	$p_{i,t}(j) = w_{H,i,t}/\vartheta_i$. Under symmetry and the aggregator normalization this also equals $P_{i,t}$.
$Y_{i,t}, Y_i(s)$	Final-good output	$Y_{i,t} = F_i(A_{X,i,t}X_{i,t}, A_{L,i,t}L_i) = e_{i,t} + \mathcal{D}_{i,t} - \mathcal{I}_{i,t}$.
$F_{i,X,t}, F_{i,L,t}$	Marginal products evaluated at the equilibrium effective inputs	Partial derivatives of primitive F_i at $(A_{X,i,t}X_{i,t}, A_{L,i,t}L_i)$. The functions $F_{i,X}, F_{i,L}$ follow from technology; their evaluated values depend on the state and allocation.
$P_{i,t}, P_i(s)$	Price of the knowledge-intensive composite good	$P_{i,t} = F_{i,X,t}A_{X,i,t}$.
$w_{H,i,t}, w_{L,i,t}, w_{a,i,t}$	Skilled wage, unskilled wage, and type-specific young income	$w_{H,i,t} = \vartheta_i F_{i,X,t}A_{X,i,t}$; $w_{L,i,t} = F_{i,L,t}A_{L,i,t}$; $w_{a,i,t}$ selects H or L . R&D workers receive the same skilled income through entry revenues when R&D is active.
$q_{i,t}(j), q_{i,t}, q_i(s)$	Ex-dividend price of one firm-level equity claim	Identical claims have $q_{i,t}(j) = q_{i,t}$. Free entry implies $q_{i,t} \leq w_{H,i,t}/(a_i N_{i,t})$, with equality only when $\varphi_{i,t} < 1$. At zero R&D, equity-market conditions and the inequality jointly determine price.
$d_{i,t}(j), d_{i,t}, d_i(s)$	Dividend / monopoly profit per incumbent variety	$d_{i,t}(j) = d_{i,t} = \frac{1 - \vartheta_i}{\vartheta_i} w_{H,i,t} x_{i,t} = \frac{1 - \vartheta_i}{\vartheta_i} w_{H,i,t} \frac{\varphi_{i,t} H_i}{N_{i,t}}$.
$\mathcal{Q}_{i,t}, \mathcal{Q}_i(s)$	Country equity-market capitalization after current issuance	$\mathcal{Q}_{i,t} = N_{i,t+1} q_{i,t}$. It is an aggregate accounting value, not the primitive per-variety price and not $N_{i,t+1} q_{i,t+1}$.
$\mathcal{D}_{i,t}, \mathcal{D}_i(s)$	Aggregate dividends paid by incumbent firms	$\mathcal{D}_{i,t} = N_{i,t} d_{i,t} = \frac{1 - \vartheta_i}{\vartheta_i} w_{H,i,t} X_{i,t}$.

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Symbol	Economic meaning and role	Definition or relationship to more primitive quantities
$e_{i,t}, e_i(s)$	Aggregate young-agent budget income	$e_{i,t} = H_i w_{H,i,t} + L_i w_{L,i,t}$. Endogenous labor/IPO income; it is not an exogenous endowment and generally differs from $Y_{i,t}$.
$\mathcal{I}_{i,t}$	IPO transfer to current R&D workers	$\mathcal{I}_{i,t} = (1 - \varphi_{i,t}) H_i w_{H,i,t} = (N_{i,t+1} - N_{i,t}) q_{i,t}$. Complementarity supports this identity also at zero R&D, when both sides are zero.
$d_{i,t}/q_{i,t}$	Per-variety dividend yield	When R&D is active, $d_{i,t}/q_{i,t} = a_i(1 - \vartheta_i) \varphi_{i,t} H_i / \vartheta_i$. At the zero-R&D boundary use the ratio of equilibrium dividend and price directly.
$\mathcal{D}_{i,t}/\mathcal{Q}_{i,t}$	Aggregate dividend yield	$\mathcal{D}_{i,t}/\mathcal{Q}_{i,t} = (N_{i,t}/N_{i,t+1})(d_{i,t}/q_{i,t})$; differs from the per-variety yield because current capitalization includes newly issued claims.
$A_{X,i,t}/A_{L,i,t}, w_{L,i,t}/w_{H,i,t}, L_i w_{L,i,t}/(H_i w_{H,i,t})$	Derived relative productivity, wage, and wage-bill quantities	All are ratios of the preceding objects. In US regime u , $A_X/A_L = (\bar{A}_{X,US,u}/\bar{A}_{L,US,u}) N_t^{\xi_u - \nu_u}$ and $L_{US} w_{L,US,t}^u / (H_{US} w_{H,US,t}^u) = K_u N_t^{\psi_{US}} \varphi_t^{\rho_{US}}$.

3.2 Household choices and their aggregation

An asterisk identifies the **RoW investor**, whereas the subscript US or W on an asset identifies its **issuer**. Thus $n_{US,t}^*$ is RoW ownership of US equity. Aggregate quantities sum per-agent choices using the corresponding country masses H_i, L_i .

Symbol	Economic meaning and role	Definition or relationship to more primitive quantities
$c_{y,a,t}, c_{y,a,t}^*$	Young consumption of a US or RoW type- a agent	$c_{y,a,t} = (1 - \beta) w_{a,US,t}; c_{y,a,t}^* = \frac{1 - \beta}{1 + \chi} w_{a,W,t}$.
$c_{o,a,t+1}, c_{o,a,t+1}^*$	Old consumption of the agent born at t	US: $(q_{US,t+1} + d_{US,t+1}) n_{US,a,t} + (q_{W,t+1} + d_{W,t+1}) n_{W,a,t} + B_{a,t+1}$. RoW: the same equity-payoff formula using starred holdings, plus $B_{US,a,t+1}^* + B_{W,a,t+1}^*$. Each must be strictly positive in every successor state.
$n_{US,a,t}, n_{W,a,t}$	US type- a holdings of each country's firm-level equity claims	Nonnegative quantities purchased at t ; values sum to $S_{a,t}$. They are measures of claims, not ownership fractions or equity-market values.
$n_{US,a,t}^*, n_{W,a,t}^*$	RoW type- a holdings of each country's equity claims	Nonnegative; values sum to $S_{a,t}^*$.
$B_{a,t+1}$	US type- a position in the US one-period bond, measured by its date- $(t+1)$ payoff	May be negative subject to budgets and positive successor consumption; present value is $b_{a,t} = q_{US,t}^B B_{a,t+1}$.
$B_{US,a,t+1}^*, B_{W,a,t+1}^*$	RoW type- a bond positions	The convenience term requires $B_{US,a,t+1}^* > 0$; the RoW bond has no such convenience term.
$S_{a,t}, S_{a,t}^*$	Type-level total equity investment at purchase prices	$S_{a,t} = q_{US,t} n_{US,a,t} + q_{W,t} n_{W,a,t}; S_{a,t}^* = q_{US,t} n_{US,a,t}^* + q_{W,t} n_{W,a,t}^*$.
$b_{a,t}$	Present value of the US type- a bond position	$b_{a,t} = q_{US,t}^B B_{a,t+1}$.

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Symbol	Economic meaning and role	Definition or relationship to more primitive quantities
$A_{a,t}, A_{a,t}^*$	Type-level total saving, including bonds	$A_{a,t} = S_{a,t} + b_{a,t} = \beta w_{a,US,t}; A_{a,t}^* = S_{a,t}^* + q_{US,t}^B B_{US,a,t+1}^* + q_{W,t}^B B_{W,a,t+1}^* = \frac{\beta + \chi}{1 + \chi} w_{a,W,t}$.
$\omega_{a,t}, \omega_{a,t}^*; \omega_t, \omega_t^*$	Share of equity investment allocated to US equity	$\omega_{a,t} = q_{US,t} n_{US,a,t} / S_{a,t}$; starred analogue for RoW. Homogeneity gives common shares across types in each country: $\omega_{a,t} = \omega_t, \omega_{a,t}^* = \omega_t^*$. Both lie in $[0, 1]$.
$\theta_{a,t}; \theta_t$	US bond share of total saving	$\theta_{a,t} = b_{a,t} / A_{a,t} = \theta_t$. After market clearing, $\theta_t = -\theta_{US,t}^* A_t^* / A_t < 0$. This is a financing share, not the production share $\varphi_{i,t}$.
$x_{US,a,t}, x_{W,a,t}$	US type- a equity positions measured in current value	$x_{US,a,t} = q_{US,t} n_{US,a,t}; x_{W,a,t} = q_{W,t} n_{W,a,t}; A_{a,t} = x_{US,a,t} + x_{W,a,t} + b_{a,t}$. Distinct from intermediate-good quantities.
$n_{US,t}, n_{W,t}$	Aggregate US holdings of each country's equity	$n_{i,t} = H_{US} n_{i,H,t} + L_{US} n_{i,L,t}$. Value shares imply $q_{US,t} n_{US,t} = \omega_t S_t$ and $q_{W,t} n_{W,t} = (1 - \omega_t) S_t$.
$n_{US,t}^*, n_{W,t}^*$	Aggregate RoW holdings of each country's equity	$n_{i,t}^* = H_W n_{i,H,t}^* + L_W n_{i,L,t}^*$. Equity clearing: $n_{i,t} + n_{i,t}^* = N_{i,t+1}$.
$B_{US,t+1}$	Aggregate US position in US bonds	$B_{US,t+1} = H_{US} B_{H,t+1} + L_{US} B_{L,t+1} = -B_{US,t+1}^* < 0$. This is the bond quantity; $B_{US,t}$ is also reused elsewhere for a stock bubble.
$B_{US,t+1}^*, B_{W,t+1}^*$	Aggregate RoW bond positions	$B_{i,t+1}^* = H_W B_{i,H,t+1}^* + L_W B_{i,L,t+1}^*$. Clearing gives $B_{US,t+1}^* = -B_{US,t+1} > 0$ and $B_{W,t+1}^* = 0$. Zero is an equilibrium restriction, not an exogenous bond demand.
$b_t, b_{US,t}^*, b_{W,t}^*$	Aggregate bond positions at present value	$b_t = q_{US,t}^B B_{US,t+1}; b_{US,t}^* = q_{US,t}^B B_{US,t+1}^*; b_{W,t}^* = q_{W,t}^B B_{W,t+1}^*$. Clearing gives $b_t + b_{US,t}^* = 0, b_{W,t}^* = 0$.
$A_t, A_t^*; A(s), A^*(s)$	Aggregate total saving	$A_t = H_{US} A_{H,t} + L_{US} A_{L,t} = \beta e_{US,t}; A_t^* = \frac{\beta + \chi}{1 + \chi} e_{W,t}$. Here $A_{L,t}$ denotes an unskilled US agent's saving, not labor-augmenting productivity.
$S_t, S_t^*; S(s), S^*(s)$	Aggregate equity investment by US and RoW investors	$S_t = (1 - \theta_t) A_t = q_{US,t} n_{US,t} + q_{W,t} n_{W,t}; S_t^* = (1 - \theta_{US,t}^* - \theta_{W,t}^*) A_t^* = q_{US,t} n_{US,t}^* + q_{W,t} n_{W,t}^*$.
$\theta_{US,t}^*, \theta_{W,t}^*$	RoW bond shares of total saving	$\theta_{US,t}^* = b_{US,t}^* / A_t^*$ and $\theta_{W,t}^* = b_{W,t}^* / A_t^*$. The reduced-system domain has $0 < \theta_{US,t}^* < 1$, and clearing imposes $\theta_{W,t}^* = 0$. Common across RoW types.
$C_{y,t}, C_{y,t}^*$	Aggregate young consumption	$C_{y,t} = H_{US} c_{y,H,t} + L_{US} c_{y,L,t} = (1 - \beta) e_{US,t}; C_{y,t}^* = H_W c_{y,H,t}^* + L_W c_{y,L,t}^* = \frac{1 - \beta}{1 + \chi} e_{W,t}$.
$C_{o,t+1}, C_{o,t+1}^*; C_{o,t}, C_{o,t}^*$	Aggregate old consumption	$C_{o,t+1} = A_t R_{A,t+1}; C_{o,t+1}^* = A_t^* R_{A,t+1}^*$. The date- t versions refer to the preceding cohort.
$U_{a,t}^*$	RoW type- a lifetime utility attained at a feasible portfolio and consumption choice	$(1 - \beta) \log c_{y,a,t}^* + \frac{\beta}{1 - \gamma} \log E_t[(c_{o,a,t+1}^*)^{1-\gamma}] - \frac{\kappa}{2} (\omega_{a,t}^* - \bar{\omega}^*)^2 + \chi \log(q_{US,t}^B B_{US,a,t+1}^*)$. Its coefficients are primitive; its value is endogenous.

3.3 Asset returns, pricing kernels, and preference wedges

Symbol	Meaning and role	Definition or relationship
$q_{US,t}^B, q_{W,t}^B$	Prices of one-period bonds paying one unit next period	$q_{US,t}^B = 1/R_{f,t}$; $q_{W,t}^B = 1/R_{f,t}^W$. Their prices are determined by household Euler equations and market clearing.
$R_{f,t}, R_{f,t}^W; R_f(s), R_f^W(s)$	US and RoW gross risk-free bond returns	Equilibrium unknowns, known when portfolios are chosen at t ; not exogenous policy rates. RoW convenience utility gives $q_{US,t}^B > q_{W,t}^B$, hence $R_{f,t} < R_{f,t}^W$.
$R_{i,t+1}; R_i(s, z')$	Gross return on a per-variety equity claim	$R_{i,t+1} = (q_{i,t+1} + d_{i,t+1})/q_{i,t}$; $R_i(s, z') = [q_i(\Gamma(s, z')) + d_i(\Gamma(s, z'))]/q_i(s)$. No variety-growth factor is inserted into the return on one existing claim.
$R_{p,a,t+1}, R_{p,t+1}$	US type-level and common equity-portfolio gross return	$R_{p,a,t+1} = \omega_{a,t}R_{US,t+1} + (1 - \omega_{a,t})R_{W,t+1} = R_{p,t+1}$.
$R_{A,a,t+1}, R_{A,t+1}$	US type-level and common gross return on total saving	$R_{A,a,t+1} = (1 - \theta_{a,t})R_{p,a,t+1} + \theta_{a,t}R_{f,t} = R_{A,t+1}$. Negative θ_t represents bond-financed equity investment.
$R_{p,t+1}^*$	RoW equity-portfolio gross return	$R_{p,t+1}^* = \omega_t^*R_{US,t+1} + (1 - \omega_t^*)R_{W,t+1}$.
$R_{A,t+1}^*$	RoW gross return on total saving	$R_{A,t+1}^* = (1 - \theta_{US,t}^* - \theta_{W,t}^*)R_{p,t+1}^* + \theta_{US,t}^*R_{f,t} + \theta_{W,t}^*R_{f,t}^W$; the last term is zero in equilibrium.
$M_{t,t+1}; M(s, z'; x)$	Undistorted US stochastic discount factor	$M_{t,t+1} = R_{A,t+1}^{-\gamma}/E_t[R_{A,t+1}^{1-\gamma}]$. It depends on endogenous saving returns and the primitive transition law.
$M_{t,t+1}^*; M^*(s, z'; x)$	Undistorted RoW stochastic discount factor	$M_{t,t+1}^* = \frac{\beta}{1 - \beta} \frac{C_{y,t}^*(C_{o,t+1}^*)^{-\gamma}}{E_t[(C_{o,t+1}^*)^{1-\gamma}]} = \frac{\beta}{\beta + \chi} \frac{(R_{A,t+1}^*)^{-\gamma}}{E_t[(R_{A,t+1}^*)^{1-\gamma}]}$.
$\tilde{M}_{t,t+1}^*; \tilde{M}^*(s, z'; x)$	Normalized RoW kernel used in share FOCs	$\tilde{M}_{t,t+1}^* = \frac{\beta + \chi}{\beta} M_{t,t+1}^* = \frac{(R_{A,t+1}^*)^{-\gamma}}{E_t[(R_{A,t+1}^*)^{1-\gamma}]}$. This is not identical to the undistorted RoW SDF when $\chi > 0$.
ϕ_t	US equity-share preference wedge	$\phi_t = \frac{\kappa}{\beta(1 - \theta_t)}(\omega_t - \bar{\omega})$. Distinct from production share $\varphi_{i,t}$.
λ_t	US bond-share wedge in stock pricing	$\lambda_t = \frac{\eta\theta_t}{\beta}\Upsilon'(\theta_t)$.
Ψ_t	US effective-equity-kernel normalization	$\Psi_t = 1 - \lambda_t + \phi_t(1 - \omega_t) = E_t[M_{t,t+1}R_{US,t+1}] > 0$ when the US pricing holder has a positive US-stock position.
$\mathcal{M}_{t,t+1}$	Effective kernel pricing US equity	$\mathcal{M}_{t,t+1} = M_{t,t+1}/\Psi_t$. The generic kernel of Section 1 is held fixed for that characterization; in the production application it is this derived equilibrium object. The pricing equality requires a positive pricing-holder position.
ϕ_t^*	RoW equity-share preference wedge	$\phi_t^* = \frac{\kappa C_{y,t}^*}{(1 - \beta)S_t^*}(\omega_t^* - \bar{\omega}^*)$.
$\mathcal{M}_{t,t+1}^{*,US}, \mathcal{M}_{t,t+1}^{*,W}$	RoW effective kernels for each country's equity	$\mathcal{M}_{t,t+1}^{*,US} = M^*/[1 + \phi_t^*(1 - \omega_t^*)]$; $\mathcal{M}_{t,t+1}^{*,W} = M^*/[1 - \phi_t^*\omega_t^*]$. Their stock-pricing equalities apply at positive holdings; a zero holding instead satisfies its KKT inequality.
$\mathcal{M}_{0,t}, \mathcal{M}_{t,T}, \mathcal{M}_{\tau,T}; \mathcal{M}_{0,t-1}^u$	Products of sequential effective pricing kernels	$\mathcal{M}_{0,t} = \prod_{s=1}^t \mathcal{M}_{s-1,s}$; $\mathcal{M}_{t,T} = \prod_{s=t+1}^T \mathcal{M}_{s-1,s}$; $\mathcal{M}_{0,t-1}^u = \prod_{j=0}^{t-2} \mathcal{M}_{j,j+1}^u$. Empty products equal one, including $\mathcal{M}_{0,0} = 1$.

3.4 International accounting, fundamental values, and bubbles

Symbol	Meaning and role	Definition or relationship
NFA_t	US net foreign assets at post-trade date- t values	$NFA_t = q_{W,t}n_{W,t} + q_{US,t}^B B_{US,t+1} - q_{US,t}n_{US,t}^*$. US equity owned by RoW enters as a liability.
VA_t	Equity-price valuation component of the change in US NFA	$VA_t = n_{W,t-1}(q_{W,t} - q_{W,t-1}) - n_{US,t-1}^*(q_{US,t} - q_{US,t-1})$. Uses beginning-of-period holdings.
CA_t	Current account defined consistently with the manuscript's valuation convention	$CA_t = NFA_t - NFA_{t-1} - VA_t$. The source defines it as the residual non-valuation change; it does not supply a separate primitive current-account process.
q_t, d_t in Section 1	Generic long-lived-claim price and dividend	In the production application these correspond to the US per-variety $q_{US,t}, d_{US,t}$. They are equilibrium price/payoff objects; Section 1 itself abstracts from their economic determination.
V_t	Fundamental value of the generic claim under the selected kernel	$V_t = E_t[\sum_{s=1}^{\infty} \mathcal{M}_{t,t+s} d_{t+s}]$. This value is kernel-relative.
B_t in Section 1; $B_0; B_{US,t}$ in the aggregation bridge	Per-variety rational bubble	$B_t = q_t - V_t = \lim_{T \rightarrow \infty} E_t[\mathcal{M}_{t,T} q_T]$. $B_{US,t}$ is the US specialization. Distinct from bond face values and the locally redefined return B_t in Table 3.8.
$\mathcal{B}_t; \mathcal{B}_0$	Date-0 discounted value of the claim restricted to surviving histories	$\mathcal{B}_t = E_0[\mathcal{M}_{0,t} q_t \mathbf{1}_{\{\tau > t\}}]$. Since $\tau \geq 1$, $\mathcal{B}_0 = q_0$. Under switched-history transversality $B_0 = \lim_t \mathcal{B}_t$.
L_t^D in Section 1	Date-0 discounted dividend leakage on surviving histories	$L_t^D = E_0[\mathcal{M}_{0,t} d_t \mathbf{1}_{\{\tau > t\}}]$. In the specialized model $L_t^D / \mathcal{B}_t = d_t^u / q_t^u$.
L_t^S in Section 1	Date-0 discounted first-switch payoff leakage	$L_t^S = E_0[\mathcal{M}_{0,t}(q_t + d_t) \mathbf{1}_{\{\tau = t\}}]$. This is not the dimensionless quantity later given the same symbol in the direct-leakage proof.
a_t	Total leakage relative to the surviving discounted value	$a_t = (L_t^D + L_t^S) / \mathcal{B}_t$. In the production specialization: $a_t = d_t^u / q_t^u + \frac{1-\pi}{\pi} (C_t^u / C_t^b)^\gamma (q_t^b + d_t^b) / q_t^u$. Distinct from R&D productivity a_i .
$\mathcal{B}_{US,t}^{\text{agg}}$	Aggregate US stock bubble under the selected effective kernel	$\mathcal{B}_{US,t}^{\text{agg}} = N_{US,t+1} B_{US,t}$. Different from the surviving terminal-value object $\mathcal{B}_t$.

3.5 Markov representations and reduced equilibrium system

These are functions or vectors of the variables above, not additional independent unknowns beyond the model's equilibrium system.

Symbol	Meaning and role	Definition or relationship
$\Gamma(s, z')$	Endogenous full-state transition map	$\Gamma(s, z') = (z', [1 + a_{US}(1 - \varphi_{US}(s))H_{US}]N_{US}, [1 + a_W(1 - \varphi_W(s))H_W]N_W)$; $s_{t+1} = \Gamma(s_t, z_{t+1})$. Only the regime transition law is exogenous.

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Symbol	Meaning and role	Definition or relationship
$N_i^+(s, \varphi_i), N_i^+(s, x), N_i^+(s)$	Post-issuance supply of country- i claims	$N_i^+ = [1 + a_i(1 - \varphi_i)H_i]N_i$, evaluated at a candidate or equilibrium production share. Equivalent to $N_{i,t+1}$ on a realized path.
$\mathcal{P}(s)$	Vector of price functions	$\mathcal{P}(s) = \{P_i(s), w_{H,i}(s), w_{L,i}(s), q_i(s)\}_{i \in \{US, W\}}$. Bond returns are listed separately in the equilibrium definition.
$x(s)$; candidate vector x	Reduced vector of equilibrium allocations and bond returns	$x(s) = (\varphi_{US}(s), \varphi_W(s), \omega(s), \theta_{US}^*(s), \omega^*(s), R_f(s), R_f^W(s))$. The first five are allocation/share choices; the last two are equilibrium returns.
$X_i(s, \varphi_i), w_{H,i}(s, \varphi_i), q_i(s, \varphi_i), d_i(s, \varphi_i)$	Production and price formulas evaluated at a candidate labor allocation	Same definitions as Table 3.1. In particular, $q_i(s, \varphi_i) = w_{H,i}(s, \varphi_i)/(a_i N_i)$ is the active-R&D substitution , not an unconditional formula at $\varphi_i = 1$.
$A(s, x), A^*(s, x), S(s, x), S^*(s, x)$	Candidate aggregate saving and equity-investment values	$A = \beta e_{US}(s, \varphi_{US})$; $A^* = \frac{\beta + \chi}{1 + \chi} e_W(s, \varphi_W)$; $\theta = -\theta_{US}^* A^*/A$; $S = (1 - \theta)A$; $S^* = (1 - \theta_{US}^*)A^*$ after eliminating $\theta_W^* = 0$.
$\hat{Q}_{US}(s, x), \hat{Q}_W(s, x)$	Capitalizations implied by candidate equity demand	$\hat{Q}_{US} = \omega S + \omega^* S^*$; $\hat{Q}_W = (1 - \omega)S + (1 - \omega^*)S^*$. At an equilibrium these equal $N_i^+ q_i = Q_i$. The hat indicates a candidate demand value, not a separate asset price.
$G_{US}(s, x), G_W(s, x)$	Equity-share optimality residuals	$G_{US} = \beta(1 - \theta)E_s[M(R_{US} - R_W)] - \kappa(\omega - \bar{\omega})$; $G_W = \beta(1 - \theta_{US}^*)E_s[\tilde{M}^*(R_{US} - R_W)] - \kappa(\omega^* - \bar{\omega}^*)$. Residual is zero at an interior share, ≤ 0 at zero, and ≥ 0 at one.
$R_p(s, z'; x), R_p^*(s, z'; x), R_A(s, z'; x), R_A^*(s, z'; x)$	Candidate Markov return functions	The portfolio-return formulas of Table 3.3 evaluated at s, x, z' ; $R_A^* = (1 - \theta_{US}^*)R_p^* + \theta_{US}^* R_f$ after bond clearing. Kernels $M, M^*, \tilde{M}^*$ are recomputed from these returns.

3.6 Branch values, relative scales, and asymptotic allocations

Symbol	Meaning and role	Definition or relationship
s_t^u, s_t^b, s_τ	Endogenous branch states and inherited switch state	From a common all- u predecessor, $s_t^z = \Gamma(s_{t-1}^u, z)$. Date- t knowledge stocks are identical across the two immediate successors; productivity and choices can differ with z . $s_\tau = (b, N_{US,\tau}, N_{W,\tau})$ inherits past endogenous knowledge.
$q_t^z, d_t^z, e_t^z, z \in \{u, b\}$	US per-variety price, dividend, and young income on a successor branch	$q_t^z = q_{US}(s_t^z)$; $d_t^z = d_{US}(s_t^z)$; $e_t^z = e_{US}(s_t^z)$. They are evaluations of the same equilibrium functions, not new parameters.
C_t^z, A_{t-1}^u	US old consumption on each successor branch and common prior saving	$C_t^z = A_{t-1}^u R_{A,t}^z$, with $A_{t-1}^u = \beta e_{US,t-1}^u$. C_t^z denotes old consumption here, not total contemporaneous consumption.
$R_{US,t}^z, R_{W,t}^z, R_{p,t}^z, R_{A,t}^z$	Equity, equity-portfolio, and total-saving returns across the two successors	$R_{US,t}^z = (q_t^z + d_t^z)/q_{t-1}^u$; RoW equity uses the analogous return. $R_{p,t}^z = \omega_{t-1}^u R_{US,t}^z + (1 - \omega_{t-1}^u) R_{W,t}^z$; $R_{A,t}^z = (1 - \theta_{t-1}^u) R_{p,t}^z + \theta_{t-1}^u R_{f,t-1}$. Portfolio weights are inherited from the common predecessor.
$M_{t-1,t}^z, \mathcal{M}_{t-1,t}^z$; other u, b super-scripts	Branch evaluations of pricing kernels and other equilibrium objects	Evaluate the same definitions at the relevant state. In particular, $\mathcal{M}^b/\mathcal{M}^u = M^b/M^u = (R_{A,t}^b/R_{A,t}^u)^{-\gamma} = (C_t^u/C_t^b)^\gamma$, since the prior wedge and normalization cancel.

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Symbol	Meaning and role	Definition or relationship
Λ_t^z	Total-saving return relative to US-equity return	$\Lambda_t^z = R_{A,t}^z/R_{US,t}^z$. This portfolio multiplier is endogenous and distinct from the wedge λ_t .
ζ_t	US equity funding relative to US young saving, along all- u	$\zeta_t = Q_{US,t}^u/(\beta e_{US,t}^u)$. World clearing implies $\zeta_t - 1 = [\frac{\beta+\chi}{1+\chi}e_{W,t}^u - Q_{W,t}^u]/(\beta e_{US,t}^u)$. It is jointly endogenous with production and portfolios.
r_t	RoW real scale relative to US real scale along all- u	$r_t = (N_{W,t}^u)^{\xi_w}/N_t^{\nu_u}$. It is a scale ratio, not an interest rate.
G_t	Finite-date US knowledge-growth factor along all- u	$G_t = N_{t+1}/N_t = 1 + a_{US}(1 - \varphi_t)H_{US}$. It converges to the composite parameter G_u under the tail result.
x_t in the CES bounds and tail proof	Ratio of US effective production inputs	$x_t = A_{X,US,t}^u \varphi_t H_{US} / (A_{L,US,t}^u L_{US})$. Then $Y_{US,t}^u = A_{L,US,t}^u L_{US} B(x_t)$. Distinct from $x(s)$, $x_{i,t}$, and $x_{i,a,t}$.
T_H	A finite cone-entry date for the selected all- u equilibrium path	The selection condition requires $(N_{T_H}, N_{W,T_H}^u) \in \mathcal{C}_H$. The manuscript need not choose the first such date. Under the prefix sufficient condition one can take $T_H = T^*$. This is a property/certificate of the selected path.
$F(\varphi; N, \zeta)$	Residual of the funding-adjusted HKT scalar equation	$F(\varphi; N, \zeta) = \beta \zeta [1 + K_u N^{\psi_{US}} \varphi^{\rho_{US}}] - [1/(a_{US} H_{US}) + 1 - \varphi]$. It depends on state, funding, and a candidate allocation; it is not the production function F_i .
$\varphi^{2C}(N, \zeta)$	Interior two-country scalar labor-allocation root	The unique root of $F(\varphi; N, \zeta) = 0$ on the stated large- N , near-unit-funding domain. Along the relevant equilibrium tail, $\varphi_t = \varphi^{2C}(N_t, \zeta_t)$.
$\varphi^{HKT}(N)$	HKT scalar reference allocation used for terminal closure	$\varphi^{HKT}(N) = \varphi^{2C}(N, 1)$. It is a derived root under unit funding, not a fixed production-share parameter; $\varphi_t/\varphi^{HKT}(N_t) \rightarrow 1$ under the stated conditions.

3.7 Selected balanced-growth values and limiting equilibrium constants

A bar does not generally identify a primitive parameter. Unlike the productivity coefficients $\bar{A}$, the barred objects below summarize equilibrium allocations or values. Absorbing-branch constants can depend on the inherited relative scale $N_{W,\tau}^{\xi_w}/N_{US,\tau}^{\nu_b}$ and on the selected stationary normalized equilibrium.

Symbol	Meaning and role	Definition or relationship
$\bar{\varphi}_b, \bar{\varphi}_{W,b}$	Selected absorbing-branch US and RoW production shares	On a fixed continuation, $\varphi_{US,t} = \bar{\varphi}_b$ and $\varphi_{W,t} = \bar{\varphi}_{W,b}$ for $t \geq \tau$. US selection requires $0 < \bar{\varphi}_b < 1$; RoW may have zero R&D at $\bar{\varphi}_{W,b} = 1$.
$G_{N,US,b}, G_{N,W,b}$	Selected absorbing-branch knowledge-growth factors	$G_{N,US,b} = 1 + a_{US}(1 - \bar{\varphi}_b)H_{US}$; $G_{N,W,b} = 1 + a_W(1 - \bar{\varphi}_{W,b})H_W$.
G_b, G_W	Selected absorbing-branch real growth factors	$G_b = G_{N,US,b}^{\nu_b}$; $G_W = G_{N,W,b}^{\xi_w}$. The condition $G_b = G_W$ is part of the selected stationary world equilibrium, not a pair of independently calibrated growth rates.
$\bar{Y}_{US}, \bar{e}_{US}$	Normalized US output and young income on a fixed absorbing continuation	$\bar{Y}_{US} = Y_{US,t}/N_{US,t}^{\nu_b}$; $\bar{e}_{US} = e_{US,t}/N_{US,t}^{\nu_b}$. Equivalently, $\bar{Y}_{US} = F_{US}(\bar{A}_{X,US,b} \bar{\varphi}_b H_{US}, \bar{A}_{L,US,b} L_{US})$.

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Symbol	Meaning and role	Definition or relationship
$\bar{q}_{US}, \bar{d}_{US}$	Normalized per-variety US equity price and dividend on that continuation	$\bar{q}_{US} = q_{US,t}/N_{US,t}^{\nu_b-1}$; $\bar{d}_{US} = d_{US,t}/N_{US,t}^{\nu_b-1}$. Their ratio is the equilibrium yield δ_b .
$\bar{Q}_{US}, \bar{D}_{US}$	Normalized aggregate US capitalization and dividends	$\bar{Q}_{US} = Q_{US,t}/N_{US,t}^{\nu_b} = G_{N,US,b}\bar{q}_{US}$; $\bar{D}_{US} = D_{US,t}/N_{US,t}^{\nu_b} = \bar{d}_{US}$.
$\bar{Y}_W, \bar{e}_W$	Normalized RoW output and young income on a fixed absorbing continuation	$\bar{Y}_W = Y_{W,t}/N_{W,t}^{\xi_W}$; $\bar{e}_W = e_{W,t}/N_{W,t}^{\xi_W} = \hat{e}_W(\bar{\varphi}_{W,b})$.
$\bar{q}_W, \bar{d}_W$	Normalized per-variety RoW equity price and dividend	$\bar{q}_W = q_{W,t}/N_{W,t}^{\xi_W-1}$; $\bar{d}_W = d_{W,t}/N_{W,t}^{\xi_W-1}$. Normalized price comes from the selected equilibrium; active RoW R&D is not required.
$\bar{Q}_W, \bar{D}_W$	Normalized aggregate RoW capitalization and dividends	$\bar{Q}_W = Q_{W,t}/N_{W,t}^{\xi_W} = G_{N,W,b}\bar{q}_W$; $\bar{D}_W = D_{W,t}/N_{W,t}^{\xi_W} = \bar{d}_W$.
$\delta_b, \delta_{W,b}$	Constant per-variety dividend yields on a fixed absorbing continuation	$\delta_b = \bar{d}_{US}/\bar{q}_{US} = a_{US}(1 - \vartheta_{US})\bar{\varphi}_b H_{US}/\vartheta_{US} > 0$; $\delta_{W,b} = \bar{d}_W/\bar{q}_W > 0$. The latter need not use the active-R&D formula.
$\bar{c}_\varphi$	Limit of the normalized all- u US production share	$N_t^{\psi_{US}/\rho_{US}} \varphi_t \rightarrow \bar{c}_\varphi = (D_u/(\beta K_u))^{1/\rho_{US}}$. A closed-form equilibrium limit, not a separate fitted parameter.
$\bar{Y}_u$	Limit of normalized all- u US output	$Y_{US,t}^u/N_t^{\nu_u} \rightarrow \bar{Y}_u = \bar{A}_{L,US,u} L_{US}(1 - \alpha_{US})^{-1/(\rho_{US}-1)}$.
$\bar{e}_u$	Limit of normalized all- u US young income	$e_{US,t}^u/N_t^{\nu_u} \rightarrow \bar{e}_u = \bar{Y}_u(1 + \beta/D_u)$.
$\bar{q}_u$	Limit of normalized all- u US per-variety price	$q_t^u/N_t^{\nu_u-1} \rightarrow \bar{q}_u = \beta \bar{e}_u/G_u$; aggregate capitalization obeys $Q_{US,t}^u/N_t^{\nu_u} \rightarrow \beta \bar{e}_u$.
$\bar{c}_D$	Limit of normalized all- u US dividend yield	$N_t^{\psi_{US}/\rho_{US}} d_t^u/q_t^u \rightarrow \bar{c}_D = a_{US}(1 - \vartheta_{US})H_{US}\bar{c}_\varphi/\vartheta_{US}$.

3.8 Local proof variables and notation collisions

Symbol and context	Meaning	Exact relationship and distinction
x_s in the deterministic terminal-product proof	Local dividend-yield sequence	$x_s = d_s/q_s$. It is not an intermediate quantity, portfolio value, or effective-input ratio.
X_t ; process X in the switched-history proof	Discounted equity-price process	$X_t = \mathcal{M}_{0,t} q_t$. It is not the production input $X_{i,t}$.
ϵ_t in the direct-leakage proof	Magnitude of the previous-date US borrowing share	$\epsilon_t = -\theta_{t-1}^u > 0$. Distinct from the fixed proof tolerances $\varepsilon^I, \varepsilon_H, \epsilon_0$.
P_t^z in the direct-leakage proof	Local name for the US risky-portfolio return on branch z	$P_t^z = R_{p,t}^z$. It is not the knowledge-good price $P_{i,t}$.

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Symbol and context	Meaning	Exact relationship and distinction
F_t in the direct-leakage proof	Local name for the prior-date US risk-free return	$F_t = R_{f,t-1}$. Distinct from the production functions F_i and scalar residual $F(\varphi; N, \zeta)$.
U_t, B_t in the direct-leakage proof	Local names for continuation- and switch-state total-saving returns	$U_t = R_{A,t}^u = (1 + \epsilon_t)P_t^u - \epsilon_t F_t$; $B_t = R_{A,t}^b = (1 + \epsilon_t)P_t^b - \epsilon_t F_t$. Here B_t is neither a bond quantity nor the rational bubble, and U_t is not lifetime utility.
L_t^S in the direct-leakage proof	Dimensionless switch-leakage factor with the hazard ratio omitted	$L_t^S = (C_t^u/C_t^b)^\gamma (q_t^b + d_t^b)/q_t^u$. To reconcile the reused notation: $(L_t^S \text{ of Section 1})/\mathcal{B}_t = [(1 - \pi)/\pi](L_t^S \text{ of this proof})$.

4 Indices, sets, and operators

These are notation rather than additional parameters or economic variables. They are included to make the inventory complete without assigning index labels an economic classification.

Notation	Interpretation
$i \in \{US, W\}; US, W$	Country/issuer index and country labels; W denotes the rest of the world.
$a \in \mathfrak{T} = \{H, L\}; H, L$	Household-type index and type labels. The index a is distinct from R&D productivity a_i and leakage a_t .
$j \in [0, N_{i,t}]$	Intermediate-variety/claim index; the integration domain's measure is endogenous. The j in a kernel product is instead a dummy time index.
t, s, T, n in sums, products, and stopping arguments	Date, summation, finite horizon, and truncation indices. The time index s is distinct from the aggregate state s . T_H, T_N, T_r, T^* have specific roles recorded above.
$u, b; z, z', \tilde{z}$	Regime labels and current/successor/dummy regime arguments. In a conditional expectation, $z', \tilde{z}$ range over possible exogenous realizations. A superscript b is distinct from the bond-value variable b_t .
$\mathcal{S} = \{u, b\} \times \mathbb{R}_{++}^2$	Domain of the mixed aggregate state. The set is specified by the model; a realized point s_t contains endogenous knowledge stocks.
$\tilde{\omega}, \tilde{\omega}^* \in [0, 1]$	Arbitrary comparison allocations in the equity-share variational inequalities; not separate equilibrium shares or shocks.
X, L in $F_i(X, L)$; x, φ, N, M, ζ as function/set arguments	Dummy arguments. In particular, M in the cone's pair (N, M) denotes a candidate RoW knowledge stock, whereas $M_{t,t+1}$ denotes an SDF. Actual paths/evaluations are classified above.
$E_t, E_s, \Pr$	Conditional expectation and probability under the specified regime law; $E_s[f(z')] = \sum_{z' \in \{u, b\}} \Pi(z, z') f(z')$. They are operators, not economic variables.
$\lceil x \rceil_+$	Nonnegative ceiling: $\max\{0, \lceil x \rceil\}$. Used to construct sufficient prefix horizons.
$O(\cdot), \sim, \propto, \gtrsim, \tau \wedge n$	Asymptotic order, asymptotic equivalence, proportionality, a lower bound up to a positive constant, and $\min\{\tau, n\}$; none is an additional model object.

5 Implications for a calibration implementation

Implementation distinction	Consequence
Primitive inputs versus derived coefficients	Specify the entries in Table 1.1 and the initial conditions; calculate Table 1.2 mechanically. Do not independently target both a primitive parameter combination and the same composite coefficient without an explicit identifying restriction.
Endogenous state versus exogenous shock	The regime follows II, but knowledge follows equilibrium R&D. Productivity levels inherit both sources of variation through the primitive maps.
Equity price versus capitalization	Store per-variety $q_{i,t}$ separately from $Q_{i,t} = N_{i,t+1}q_{i,t}$; use incumbent $N_{i,t}$ for dividends.
Interior R&D versus no-R&D boundary	Apply $q_{i,t} = w_{H,i,t}/(a_i N_{i,t})$ only where $\varphi_{i,t} < 1$. Else retain the inequality and complementarity condition.
Reduced equilibrium unknowns	The source's reduced vector contains seven coordinates: $(\varphi_{US}, \varphi_W, \omega, \theta_{US}^*, \omega^*, R_f, R_f^W)$. Recover $\theta_W^* = 0$ and $\theta = -\theta_{US}^* A^*/A$ from bond clearing, with all returns and continuation functions evaluated consistently.
Constant equilibrium quantities versus primitive parameters	Solve for absorbing shares, normalized valuations, and growth rates under the selected continuation. Their constancy on that continuation does not turn them into independent primitives.
Structural restrictions versus sufficient-theorem restrictions	$\xi_u > \nu_u > \nu_b \geq 0$, $\rho_{US} > 1$, $0 < \gamma < 1$, and $(1 + a_{US}H_{US})^{\nu_u} > (1 + a_W H_W)^{\xi_w}$ belong to the sufficient bubble argument. The absorbing common-growth condition and prefix funding bounds additionally restrict the equilibrium selection; they are not a complete primitive characterization of bubble existence.
Source domain gaps	The source does not give a complete explicit domain for κ , η , or $\bar{\omega}^*$. Any numerical implementation must state its chosen restrictions rather than attribute them to the manuscript.